

BUSINESS REQUIREMENTS DOCUMENT

U.S. Medicaid Drug Utilization & Cost Intelligence

A 10-Year State-Level Analysis (2016–2024)

Document Title	Business Requirements Document — U.S. Medicaid Drug Utilization & Cost Intelligence
Project Reference	Medicaid Drug Spend Intelligence & Cost Optimization
Prepared By	Bhumi Nagar — Business Analyst
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Primary Stakeholder	Medicaid Managed Care Organization (MCO) & Payer Analytics Team
Status	Completed

1. EXECUTIVE SUMMARY

This Business Requirements Document defines the analytical and reporting requirements for the Medicaid Drug Spend Intelligence & Cost Optimization project. The project was initiated to address a critical visibility gap faced by Medicaid managed care organizations (MCOs). Despite managing billions of dollars in pharmaceutical expenditure annually, most payer analytics teams lack the structured analytical infrastructure to understand what is actually driving those costs.

This document translates stakeholder business needs into functional requirements, KPI definitions, data requirements, and acceptance criteria. It serves as the foundational reference for all analytical deliverables — data pipeline, KPI models, Power BI dashboard, and executive recommendations.

2. BUSINESS PROBLEM STATEMENT

Healthcare reimbursement systems generate massive volumes of utilization and spending data. However, Medicaid managed care organizations and payer analytics often lack centralized visibility into:

- Long-term pharmaceutical spending growth trends
- High-cost drugs are driving the majority of expenditure
- Geographic concentration of spending across states
- Comparative reimbursement model structures and efficiency
- Cost concentration risks and intervention opportunities

Without structured analytical reporting, decision-makers face challenges identifying cost drivers, prioritizing optimization efforts, and understanding utilization behavior. This project addresses the problem by building a business intelligence solution that converts raw Medicaid utilization data into actionable business insights.

3. BUSINESS OBJECTIVES

The following business objectives define the outcomes this project must achieve for the MCO payer analytics team:

- Deliver structured visibility into Medicaid pharmaceutical spending patterns across drugs, states, and time periods
- Identify the top cost-driving drugs using Pareto analysis to support formulary management and rebate negotiation
- Map 10-year spending trends to identify demand inflection points and unsustainable cost trajectories
- Benchmark state-level performance to surface geographic outliers and high-cost markets

5. Decompose cost drivers to distinguish price-driven increases from volume-driven utilization growth
6. Compare MCOU vs FFSU reimbursement model efficiency to inform payer strategy
7. Deliver actionable, business-ready recommendations the MCO analytics team can act on directly

4. CURRENT STATE ANALYSIS

Before this project, the MCO payer analytics team faced the following operational and reporting challenges:

Challenge	Business Impact
Fragmented utilization data across multiple yearly files	No consolidated view of multi-year spending patterns — trend analysis impossible
No centralized KPI reporting framework	Decision-makers lacked standard metrics to monitor cost and utilization performance
Limited visibility into cost concentration	Unable to identify which drugs drive the majority of spend — intervention opportunities missed
No reimbursement model comparison	MCOU vs FFSU efficiency unknown — payer strategy decisions made without comparative data
Inconsistent geographic spend interpretation	State-level outliers unidentified — budget planning lacked geographic context
Suppressed records unmanaged	Data quality risks unaddressed — suppressed values potentially distorting calculations

5. STAKEHOLDER NEEDS ANALYSIS

The following table captures what each stakeholder group needs from this project — translated from business goals into analytical requirements:

Stakeholder	Business Need	Analytical Requirement
MCO Payer Analytics Team	Understand which drugs are driving costs and where to intervene	Drug-level spend analysis, Pareto concentration, cost-per-prescription benchmarking
Medicaid Program Director	Monitor program-level spending trends and ensure budget sustainability	10-year trend analysis, YoY growth metrics, executive KPI dashboard
Finance & Operations Team	Understand geographic cost concentration for budget allocation planning	State-level spend comparison, cost-per-prescription by state, geographic outlier identification
Pharma Market Access Team	Understand Medicaid reimbursement model structure and drug positioning	MCOU vs FFSU comparison, drug-level spend share, formulary positioning insights

6. KEY BUSINESS QUESTIONS

This project is designed to answer the following business questions for the MCO payer analytics team:

#	Business Question	Analysis Type
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1	Which drugs are driving the majority of Medicaid pharmaceutical spend?	Pareto / Drug-Level Analysis
2	How has total Medicaid drug spend trended over the past 10 years?	Trend Analysis (YoY)
3	Which states contribute disproportionately to overall pharmaceutical expenditure?	Geographic / State Benchmarking
4	Are cost increases driven by price inflation or increased utilization volume?	Cost Driver Decomposition
5	Which reimbursement model — MCOU or FFSU — delivers better cost efficiency?	Comparative Analysis
6	How concentrated is Medicaid drug spend — do a few drugs dominate most expenditure?	Concentration / Pareto Analysis
7	What percentage of records are suppressed and how does this affect data reliability?	Data Quality / Suppression Analysis

7. PROJECT SCOPE

☑ In Scope	✗ Out of Scope
10 years CMS Medicaid data (2016–2024) - All 50 U.S. states + jurisdictions - Both utilization types: MCOU and FFSU - Drug-level, state-level, year-level analysis - KPI design, data pipeline, dashboard delivery - Business recommendations for cost optimization	- Patient-level or beneficiary-level data - Clinical efficacy or therapeutic outcome analysis - Real-time or live data feeds - Predictive modeling or forecasting - Provider-level patient analysis - Production cloud deployment infrastructure

8. FUNCTIONAL REQUIREMENTS

The following functional requirements define what the analytical solution MUST deliver. Each requirement is assigned a unique FR ID for traceability.

FR-01: Data Ingestion Requirements

FR ID	Requirement
FR-001	System shall import yearly Medicaid utilization CSV files into PostgreSQL for all years 2016–2024
FR-002	System shall validate imported row counts and year coverage after each file import
FR-003	System shall support multi-year analytical consolidation within a single unified schema

FR-02: Data Transformation Requirements

FR ID	Requirement
FR-004	Systems should clean and standardize source data, including column formats and naming conventions

FR-005	System shall handle suppressed records by applying NULL values to spend and utilization fields where suppression_used = TRUE
FR-006	System shall generate reusable staging views separating raw, clean, and enriched analytical layers
FR-007	System shall exclude invalid geographic values ('XX') from state-level analysis and document exclusions
FR-008	System shall engineer derive KPI metrics: Cost per Prescription, Cost per Unit, Units per Prescription, Medicaid Dependency Ratio

FR-03: KPI Modeling Requirements

FR ID	Requirement
FR-009	System shall generate year-level KPI summaries covering spend, prescriptions, units, and derived metrics
FR-010	System shall generate state-level KPI analysis with geographic benchmarking capability
FR-011	System shall generate drug-level spend analysis ranked by total reimbursement
FR-012	System shall implement Pareto analysis using cumulative spend share to identify 80/20 cost concentration
FR-013	System shall compare MCOU and FFSU reimbursement models across all KPI dimensions
FR-014	System shall produce suppression analysis at year, state, and utilization type levels

FR-04: Dashboard Requirements

FR ID	Requirement
FR-015	Dashboard shall display executive KPI summary cards: Total Spend, Prescriptions, Units, Cost/Rx, Medicaid Dependency %
FR-016	Dashboard shall display 10-year spending trend visualization — 2025 excluded due to partial-year data
FR-017	Dashboard shall support MCOU vs FFSU payment model comparison with side-by-side KPI analysis
FR-018	Dashboard shall display the Top 10 drugs by spend and the Pareto concentration chart
FR-019	Dashboard shall display state-level geographic concentration analysis across all 50+ states
FR-020	Dashboard shall include an executive insight summary narrative on each page
FR-021	Dashboard shall include a Data Quality page documenting suppression rate by year, state, and utilization type

9. NON-FUNCTIONAL REQUIREMENTS

The following non-functional requirements define the quality, usability, and performance standards the solution must meet:

NFR ID	Requirement	Rationale
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NFR-001	Dashboards shall maintain clear executive readability without requiring technical knowledge	MCO decision-makers are business users — not data engineers
NFR-002	All visualizations shall support business interpretation — findings framed as decisions, not just data	Business value comes from actionable insights, not raw numbers
NFR-003	KPI calculations shall remain consistent and reproducible across all dashboard pages	Inconsistent metrics undermine stakeholder trust in the analysis
NFR-004	Data transformations shall be modular and reusable — structured for future analytical expansion	Supports scalability for additional years, states, or drug categories
NFR-005	Dashboard design shall use a consistent healthcare-appropriate visual theme throughout	Professional presentation supports executive and portfolio credibility

10. DATA REQUIREMENTS

Field	Data Type	Business Purpose	Notes
Year	Integer	Trend analysis, YoY comparisons	2025 excluded — partial year
State	Text	Geographic benchmarking	'XX' records excluded
Product Name	Text	Drug-level spend analysis	Product-name level only — no therapeutic class
Utilization Type	Text	MCOU vs FFSU comparison	MCOU = Managed Care, FFSU = Fee-for-Service
Total Amount Reimbursed	Numeric	Primary spend metric	NULL when suppression_used = TRUE
Medicaid Amount Reimbursed	Numeric	Government spend dependency	Used for Medicaid Dependency Ratio
Non-Medicaid Amount Reimbursed	Numeric	Non-government spend split	Complement to Medicaid spending
Number of Prescriptions	Numeric	Utilization volume baseline	Used in Cost per Prescription
Units Reimbursed	Numeric	Dosage intensity analysis	Used in Cost/Unit, Units per Rx
Suppression Used	Boolean	Data quality governance	TRUE = record withheld for privacy

11. KPI DEFINITIONS

The following KPIs were designed to answer specific business questions for the MCO payer analytics team. Each KPI includes its business purpose, calculation method, and source table.

KPI Name	Business Purpose	Formula	Source Table
Total Spend	Track overall cost burden across time, state, drug	SUM(total_amount_reimbursed)	kpi_yearly, kpi_state, kpi_drug
Medicaid Spend	Isolate the government-funded portion of the spend	SUM(medicaid_amount_reimbursed)	kpi_yearly, kpi_state
Total Prescriptions	Measure utilization volume baseline	SUM(number_of_prescriptions)	kpi_yearly, kpi_drug
Avg Cost per Prescription	Assess pricing pressure per dispensing event	Total Spend / Total Prescriptions	All KPI views
Avg Cost per Unit	Evaluate per-unit pricing efficiency	Total Spend / Units Reimbursed	All KPI views
Avg Units per Prescription	Understand dosage intensity per dispensing	Units Reimbursed / Prescriptions	All KPI views
Medicaid Dependency Ratio	Measure the government vs. another payer split	Medicaid Spend / Total Spend	All KPI views
Spend Share	Calculate each drug's % of total spend	Drug Spend / SUM(All Drug Spend)	kpi_drug_share
Cumulative Spend Share	Enable Pareto 80/20 concentration analysis	Running SUM of Spend Share (DESC)	kpi_drug_pareto
Suppression Rate	Assess data completeness and reliability	Suppressed Rows / Total Rows	kpi_suppression_year/state

12. ANALYTICAL METHODS USED

Method	Business Purpose	Applied To
Trend Analysis	Identify spending growth patterns over 10 years	Year-level KPIs — spend, prescriptions, units
Pareto Analysis	Identify 80/20 cost concentration in drug spend	kpi_drug_pareto — cumulative spend share
Comparative Analysis	Compare MCOU vs FFSU efficiency across KPIs	kpi_utilization_type

Geographic Segmentation	Benchmark state-level spending and cost efficiency	kpi_state, kpi_state_year
Cost Driver Decomposition	Separate price vs. volume contribution to cost growth	Cost/Rx vs prescription volume trend analysis
Suppression Analysis	Assess data quality and reporting completeness	kpi_suppression_year/state/utilization
Feature Engineering	Create derived business metrics, not in raw data	Cost/Rx, Cost/Unit, Units/Rx, Dependency Ratio

13. ACCEPTANCE CRITERIA

The project deliverables will be considered complete and accepted when ALL the following criteria are met:

#	Acceptance Criterion	Verification Method
1	All 10 years of data (2016–2024) successfully loaded and validated in PostgreSQL	Row count validation by year
2	All KPI views produce non-null results for non-suppressed records	SQL validation queries on each view
3	Pareto analysis correctly identifies cumulative spend share at 80% threshold	kpi_drug_pareto — WHERE cumulative_spend_share <= 0.80
4	Power BI dashboard loads all KPI tables without errors	Dashboard refresh test
5	The Executive Overview page answers all core business questions briefly	Stakeholder review of page 1
6	The MCOU vs FFSU comparison clearly shows which model delivers better cost efficiency	Side-by-side KPI comparison visual
7	Minimum 5 business recommendations delivered with supporting data and evidence	Executive Summary document review
8	Suppression rates documented by year, state, and utilization type	Data Quality dashboard page review

14. ASSUMPTIONS & CONSTRAINTS

Assumptions

- CMS published Medicaid data is accurate and representative of actual state drug utilization patterns
- Suppressed records represent low-volume data withheld for privacy — excluded from spend calculations
- MCOU and FFSU represent the two primary reimbursement model categories available in the dataset
- Product names are sufficiently standardized for aggregation at the product level
- 2025 data excluded from trend analysis due to partial-year reporting — documented in data quality notes

Constraints

- No patient-level or beneficiary identifiers available — analysis at aggregated state-drug level only
- Therapeutic drug classification not available in source data — analysis limited to product name level
- Invalid geographic labels ('XX') excluded from state analysis — documented as unclassified jurisdictions
- Historical analysis only — no real-time or live data feed available

- Drug spend concentration pattern may not reflect future formulary changes or rebate adjustments

15. RISKS & MITIGATION

Risk	Impact	Mitigation
Partial-year 2025 data	Misleading trend	2025 excluded from executive trend visuals — documented in data quality notes
Suppressed data records	Incomplete calculations	Suppression handled via conditional logic — NULLs applied; tracked via separate KPI views
Invalid geographic labels ('XX')	Distorted state analysis	'XX' records excluded from geographic visuals — documented as unclassified jurisdictions
Decimal overflow in Power BI	Import instability	Isolated to specific high-precision records — managed through controlled data type formatting

16. DOCUMENT SIGN-OFF

This Business Requirements Document has been prepared by Bhumi Nagar and defines the complete analytical and reporting requirements for the Medicaid Drug Spend Intelligence project. All deliverables — data pipeline, KPI models, Power BI dashboard, and executive summary — will be developed in alignment with the requirements defined in this document.

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